Earth, Sun and Cyclic Phenomena

YEAR 6 SCIENCE | AC9S6U02

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Analyse and Practise

Name: ______________________________ Date: ______________

LEARNING INTENTION	model how Earth's rotation, revolution and tilt relate to observable cycles
KEY IDEA	Earth rotates on its axis about once each day and revolves around the sun about once each year. Its tilted axis changes sun angle and day length through the year; seasons are not caused by distance from the sun.
WORKED MODEL	When a location rotates toward the sun it experiences daylight. Six months later, the same hemisphere may tilt more directly toward the sun, producing longer days and more concentrated sunlight.

SUCCESS CRITERIA - I CAN

[ ] distinguish rotation/revolution [ ] use a model [ ] explain day length and seasons [ ] correct misconceptions

A. NOTICE AND IDENTIFY

1. Match rotation, revolution and axial tilt to day/night, year and seasonal day-length change.

2. Label sun, Earth, axis, illuminated half and direction of rotation on a model.

3. Explain why half of Earth is always lit even though a location experiences night.

B. APPLY AND EXPLAIN

4. Predict what happens to a shadow from morning to midday to afternoon.

5. Compare summer and winter day length in the same Australian location.

6. Correct: Australia has summer because Earth is much closer to the sun in January.

C. INDEPENDENT FLUENCY - CHECK AND CREATE

7. Draw four Earth positions around the sun with parallel tilted axes and label one hemisphere's seasons.

8. Check the model's strengths and one limitation of scale.

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Name: ______________________________ Date: ______________

D. APPLY THE SKILL IN CONTEXT

9. A globe and lamp model is used. Write a safe method to demonstrate day/night and seasons.

10. Explain observations using rotation, tilt, sun angle and duration of daylight.

11. Predict how patterns differ near the equator and at high southern latitudes.

E. REASON, EVALUATE AND TRANSFER

12. Diagnose a diagram whose axis changes direction at every orbital position.

13. Create a five-sentence explanation linking daily and yearly cycles without using the distance misconception.

F. STUDENT REVIEW - PROVE YOUR UNDERSTANDING

[ ] I can model and explain how Earth-sun relationships create observable cycles.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: ____

The most difficult decision was __________________ because _______________________________

One correction or improvement I made: ____

My next practice step: ____

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Accept equivalent responses when the student applies the targeted skill accurately and justifies the choice.

COMPLETE ANSWER KEY

1. rotation-day/night; revolution-year; tilt-seasonal day length/sun angle.

2. Labels must show fixed axis orientation and one illuminated hemisphere.

3. Earth is spherical; the sun-facing half is lit while rotation moves locations through light/dark.

4. Shadow generally shortens toward solar noon then lengthens, changing direction.

5. Australian summer generally has longer daylight; winter shorter.

6. Seasons result mainly from axial tilt, sun angle and daylight duration.

7-8. Axes remain parallel; model is not to scale and lamp rays/size differ from reality.

9. Rotate globe for day/night; move it around lamp with constant tilt; avoid hot lamp contact.

10. Explanation must connect all four terms causally.

11. Equator varies less; high latitudes vary more.

12. Keep axis pointing in the same direction throughout orbit.

13. Require accurate daily/yearly distinction and no distance cause.

TEACHER CHECK AND NEXT STEP

[ ] Uses accurate science ideas and evidence.

[ ] Explains a decision rather than listing an answer.

Next step: model one difficult item, complete one together, then transfer the skill to a new investigation.

COMMON MISCONCEPTION / WATCH OUT

Seasons are not caused by Earth being nearer or farther from the sun.

The axis keeps nearly the same orientation as Earth revolves.

EXTENSION AND RE-TEACH

Re-teach: model one item aloud, complete one together, then ask the student to explain an independent correction.

Extension: transfer the skill to a short authentic text and justify two deliberate language choices with quoted evidence.